AQ7.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
Choose the set of correct options.

☐ If a $3 \times 33 \times 3$ matrix AA is similar to the identity matrix of order 33, then AA must be the identity matrix of order 3.

☐ If a $3 \times 33 \times 3$ matrix AA is similar to a diagonal matrix of order 33, then AA must be a diagonal matrix of order 3.

☐ If AA and BB are similar matrices, then they are also equivalent matrices.

☐ If AA and BB are equivalent matrices, then they are also similar matrices.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If a $3 \times 33 \times 3$ matrix AA is similar to the identity matrix of order 33, then AA must be the identity matrix of order 3.
If AA and BB are similar matrices, then they are also equivalent matrices.

Consider the linear transformation given below:
$T: R^3 \rightarrow R^2$ T : R3 → R2

$T(x, y, z) = (x - y, y + z)$T(x, y, z) = (x − y, y + z)

Consider two ordered bases $\beta_1 = \{(1,0,0),(0,1,0),(0,0,1)\}$β1 = {(1, 0, 0), (0, 1, 0), (0, 0, 1)}, and $\beta_2 = \{(1,0,1),(0,1,1),(1,1,0)\}$β2 = {(1, 0, 1), (0, 1, 1), (1, 1, 0)} of R^3 R3 and consider the standard ordered basis $\gamma = \{(1,0),(0,1)\}$γ = {(1, 0), (0, 1)} of R^2 R2. Let AA and BB be the matrices corresponding to the linear transformation TT with respect to the bases β_1 β1 and β_2 β2 for the domain, respectively and γγ for the co-domain. Let QQ and PP be matrices such that $B = QAP$B = QAP.

Answer the questions 2,3, and 4 using the above information.
1 point
Which of the following matrices represents AA?

○ $\begin{bmatrix}1 & -1 & 0\\ 1 & 2 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ -1 & 2\\ 0 & 1\end{bmatrix}$

○ $\begin{bmatrix}1 & -1 & 0\\ 0 & 1 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ -1 & 1\\ 0 & 1\end{bmatrix}$

○ $\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix}\begin{bmatrix}1 & -1 & 0\\ 1 & 1 & 1\end{bmatrix}$

○

$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 & 0 & 1 \\ 0 & 1 & 2 & 1 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$

1 point
Which of the following matrices can be P?

There does not exist any matrix P which satisfies the property $B = QAP$.

$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & 1 \\ 1 & 2 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}\begin{bmatrix} 2 & 1 & -2 & 1 \\ 2 & 1 & -2 & 1 \\ 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 1 \\ 2 & 1 & -2 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$

1 point
Which of the following statements is true for Q?

Q can be the identity matrix of order 3.

Q can be the identity matrix of order 2.

Whatever the matrix P we choose, Q is always unique.

Q can be the matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 & 1 & 0 \end{bmatrix}$.

No, the answer is incorrect.
Score: 0
Accepted Answers:

QQ can be the identity matrix of order 2.
QQ can be the matrix $\begin{bmatrix}0&1\\1&0\end{bmatrix}$[0110].

1 point
Choose the set of correct options.
☐ If two square matrices of the same order have the same determinants, then they must be similar to each other.
☐ Similar matrices have the same rank.
☐
If AA and BB are similar matrices, and $Ax = b$Ax = b has a unique solution, then $Bx = b$Bx = b also has a unique solution.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Similar matrices have the same rank.
If AA and BB are similar matrices, and $Ax = b$Ax = b has a unique solution, then $Bx = b$Bx = b also has a unique solution.

1 point
Choose the set of correct options.
☐
If det$A$A or det$B$B is nonzero, then ABAB and BABA are similar.
☐
$A = \begin{bmatrix}2&1\\5&3\end{bmatrix}$A = [2513] and $B = \begin{bmatrix}2&3\\1&2\end{bmatrix}$B = [2132] are similar.

☐
$A = \begin{bmatrix}2&1\\0&3\end{bmatrix}$A = [2013] and $B = \begin{bmatrix}1&11\\0&4\end{bmatrix}$B = [10114] are similar.

☐
$A = \begin{bmatrix}2&1\\0&2\end{bmatrix}$A = [2012] and $B = \begin{bmatrix}2&0\\0&2\end{bmatrix}$B = [2002] are not similar.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If det$A$A or det$B$B is nonzero, then ABAB and BABA are similar.
$A = \begin{bmatrix}2&1\\0&2\end{bmatrix}$A = [2012] and $B = \begin{bmatrix}2&0\\0&2\end{bmatrix}$B = [2002] are not similar.

1 point
Which of the following options are correct?
☐ Two invertible matrices of same order are equivalent.
☐
$A = \begin{bmatrix}17&6\\2&-1\\5&-3\end{bmatrix}$A = $\begin{bmatrix}1&2&5&7\\1&2&6&-1\\-3&&&\end{bmatrix}$ and $B = \begin{bmatrix}2&3&-2\\1&1&4\\1&2&-6\end{bmatrix}$B = $\begin{bmatrix}2&1&1&3\\1&2&-2&4\\-6&&&\end{bmatrix}$ are equivalent.

☐
$A = \begin{bmatrix}2&3\\4&6\end{bmatrix}$A = [2436] and $B = \begin{bmatrix}1&3\\3&1\end{bmatrix}$B = [1331] are equivalent.

☐

Consider two linear transformations $T_1(x, y, z) = (x + y - z, x + 4y - 2z, -2x + y + z)$T1 (x, y, z) = (x + y − z, x + 4y − 2z, −2x + y + z) and $T_2(x, y, z) = (x + y, x - z, y + z)$T2(x, y, z) = (x + y, x − z, y + z). Let AA and BB denote the matrix representation of T_1 T1 and T_2 T2 with respect to the standard basis of R^3 R3. Then AA and BB are equivalent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Two invertible matrices of same order are equivalent.

$A = \begin{bmatrix} 1 & 7 & 6 \\ 2 & 1 & -1 \\ 5 & 2 & -3 \end{bmatrix}$A = $\begin{bmatrix} 1 & 25 & 7 & -6 \\ 1 & 2 & 6 & -1 \\ -3 & -1 & -5 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 & -2 \\ 1 & 1 & 4 \\ 1 & 2 & -6 \end{bmatrix}$B = $\begin{bmatrix} 2 & 1 & 1 & 3 \\ 1 & 2 & -2 & 4 \\ -6 & -2 & 4 & -6 \end{bmatrix}$ are equivalent.

Consider two linear transformations $T_1(x, y, z) = (x + y - z, x + 4y - 2z, -2x + y + z)$T1 (x, y, z) = (x + y − z, x + 4y − 2z, −2x + y + z) and $T_2(x, y, z) = (x + y, x - z, y + z)$T2(x, y, z) = (x + y, x − z, y + z). Let AA and BB denote the matrix representation of T_1 T1 and T_2 T2 with respect to the standard basis of R^3 R3. Then AA and BB are equivalent.

Level 2

1 point

Let EE be the set of $2 \times 2$2 × 2 matrices which are similar to a scalar matrix $A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$A = [3003].

Choose the set of correct statements.

☐ EE is an infinite set.

☐ Cardinality of EE is finite.

☐ EE is singleton set.

☐ EE consists of all the scalar matrices of order nn.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Cardinality of EE is finite.

EE is singleton set.

1 point

Which of the following option(s) is(are) true?

☐ If A and BA and B are similar matrices then A^{-1} and B^{-1} A−1 and B−1 are similar matrices.

☐ If A and BA and B are similar matrices then A^2 and B^2 A2 and B2 are similar matrices.

☐ If A^2 and B^2 A2 and B2 are similar matrices then A and BA and B are similar matrices.

☐ If A and BA and B are similar matrices then A^T and B^T AT and BT are similar matrices.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If A and BA and B are similar matrices then A^{-1} and B^{-1} A−1 and B−1 are similar matrices.

If A and BA and B are similar matrices then A^2 and B^2 A2 and B2 are similar matrices.

If A and BA and B are similar matrices then A^T and B^T AT and BT are similar matrices.
1 point
Let us consider the following matrices.

$$A = \begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}, B = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}, C = \begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}\text{A} = [1101]\,, \text{B} = [1001]\,, \text{C} = [1011]$$

Choose the set of correct options.

- [ ] AA and BB have the same rank, and same determinant, but they are not similar matrices.
- [ ] AA and CC have the same rank, and same determinant, but they are not similar matrices.
- [ ] BB and CC have the same rank, and same determinant, but they are not similar matrices.
- [ ] None of the options are true.

No, the answer is incorrect.
Score: 0
Accepted Answers:
AA and BB have the same rank, and same determinant, but they are not similar matrices.
BB and CC have the same rank, and same determinant, but they are not similar matrices.

Check Answers
Your score is: 0/10